

GNSS EXPECTED RANGE

Assigned: 15 September 2026

Due: 25 September 2026

OBJECTIVES

This assignment takes the first few steps in putting together code for a least squares position solution. The three main objectives are:

- Read and interpret observation data directly from a RINEX observation file.
- Compute the expected range from the GPS satellite at the time of transmission, to the receiver at the time of reception, accounting for signal travel time and Earth rotation.
- Compare measured pseudorange with the expected geometric range.

OVERVIEW

We will work continue to work with data for one day and one location:

- Date: Wednesday, August 19, 2026 = GPS Week 2432 (or 384), TOW 259200-345600/ DOY 231
- User locations:
 - NIST (in Boulder, CO): ECEF coordinates: [-1288398.567 -4721696.932 4078625.350] (m)
- RINEX File: NIST00USA_R_20262310000_01D_30S_MO.rnx
Downloaded from CDDIS: <https://cddis.nasa.gov/archive/gnss/data/daily/>
- Broadcast ephemeris file: brdcDOY0.26n (DOY is the 3-digit day of year)
Downloaded from CDDIS: <https://cddis.nasa.gov/archive/gnss/data/daily/>
- SP3 file: IGS00PSFIN_20262310000_01D_15M_ORB.SP3
Downloaded from CDDIS. *Caution – this is in km, all other positions in this HW are in meters.*
- You will need to use gunzip and crx2rnx to uncompress the observation file:
<https://terras.qsi.go.jp/ja/crx2rnx.html>
The data files provided on CANVAS have already been uncompressed & converted to rnx, so uncompress is only required if you download the files from CDDIS yourself.

ASSIGNMENT

1) GPS Observations – We will start out using PRN 05

Download the NIST RINEX observation file for the specified date from IGS / CDDIS.

a. Use the built-in MATLAB function rinexread (or another version that you find or write yourself) to read the RINEX file and extract the GPS measurements.

b. Plot the C1C pseudorange vs time for the specified satellite. Show the time in hours into the day or in the MATLAB datetime-format given in the rinexread output.

c. Plot the signal to noise ratio S1C vs time for the specified satellite.

d. Check out one or two of the other pseudorange or carrier phase measurement types (convert carrier phase to meters) and briefly discuss how they compare to C1C.

2) GPS positions based on ephemeris data

Download the broadcast ephemeris file for the specified date from Canvas or from IGS / CDDIS.

a. Use the function `read_clean_GPSbroadcast.m` (or a similar function you find elsewhere or write yourself) to load the ephemeris data into a numerical array.

b. Use the function `eph2pvt2026` (or similar function you write or find) to compute the position of your GPS satellite based on the ephemeris data, for the times given in the `sp3` file. Plot both positions vs time and plot the difference between the ephemeris-based positions and the precise positions. Comment on these results compared to what you found in HW2 for the almanac.

c. Get the satellite clock bias based on the `a0` and `a1` parameters and plot this as a function of time in a separate graph or figure. (Note that the relativity correction is not included in the provided code.)

d. Optional – compute the relativistic correction for the GPS clock based on the satellite positions in ECEF and plot them as a function of time. Is it important to include this correction if you are trying to find your position on the Earth at the meter level?

3) GPS expected range

a. For each of the measurement times for your satellite in the observation file, compute the satellite position based on the broadcast ephemeris. Then use the `compute_azelrange` function you wrote in HW2, to find the range (`R0`) and elevation of your satellite from the NIST IGS Site. Plot each of these versus time (hours of day or datetime format).

b. Now recompute the expected range accounting for the signal travel time and coordinate frame rotation as described below (`R1`). Add this updated range to the previous range plot and on a separate graph, plot the difference. What is the largest difference you found?

4) Observed minus computed (expected) range.

a. Plot the measured C1C pseudoranges and your expected ranges (`R1`) vs time. Then plot the difference between them (`O-C`).

b. Explain what you think is likely to be the largest source of the difference between them.

Justify your theory by looking at data from another satellite, another location, or using other information in the files.

5) Repeat your calculations for any other PRN you choose and show the plots described in Part 4. Maybe consider a satellite from a different GPS block or one that is visible at different times than the one specified above.

COMPUTING THE EXPECTED RANGE:

The expected range of interest is the distance between the position of the GPS satellite at the time of transmission (T_t) and the GPS receiving antenna at the time of reception (T_r). To compute this distance, both positions must be represented in the same coordinate frame - we use the ECEF coordinate frame at the time of reception (T_r).

STEPS FOR COMPUTING EXPECTED RANGE

- 1) Compute the GPS satellite position in ECEF at T_r based on the broadcast ephemeris.
- 2) Use an *a priori* value for the receiver coordinates (R_X) to find geometric range $R = |r_{GPS} - r_{RX}|$.
- 3) Compute the time of transmission: $T_t = T_r - R/c$ (use the real speed of light, NOT $3e8$)
- 4) Compute the satellite position at T_t in ECEF at T_t based on the broadcast ephemeris.
- 5) Rotate the satellite position to ECEF at T_r . (ω_E is the rotation rate of the Earth)

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{ECEF @ t_R} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{ECEF @ t_T}$$

where

$$\phi = \omega_E(t_R - t_T)$$

- 6) Compute a new geometric range using this position for r_{GPS} , $R = |r_{GPS} - r_{RX}|$.
- 7) Repeat steps 3-6 until convergence. (If it takes more than two iterations, something is wrong).